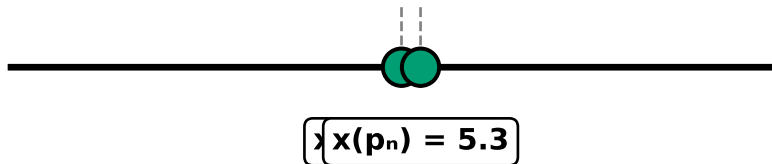


Why Full INT8 Quantization Breaks CDC: Catastrophic Cancellation Mechanism

FP32 Floating-Point (Works Correctly)

Activation values



CDC Micro-Gradient:

$$\Delta = x(p_0 + p_n) - x(p_0)$$

$$\Delta = 5.3 - 5.1 = 0.2$$

*Non-zero gradient
Backprop works ✓*

INT8 Quantized (Fails)

Quantization buckets



CDC Micro-Gradient (Quantized):

$$\Delta_q = q(x(p_0 + p_n)) - q(x(p_0))$$

$$\Delta_q = q_5 - q_5 = 0$$

*Zero gradient
Backprop blocked ✗
Model collapse*

Quantization rounding collapses nearby activations into the same bucket, destroying CDC's precision-sensitive micro-gradients